

# Henry Agyare Asante

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## EDUCATION

**Grambling State University - Grambling, Louisiana, USA**

**Graduating May 2028**

**Major:** Computer Science.

**GPA:** 4.0/4.0

**Relevant Coursework:** Data Structures, Operating Systems, Deep Learning, Databases, Calculus, Artificial Intelligence

## WORK EXPERIENCE

### Allstate, Software Engineer Intern

**May 2026 - Aug 2026**

- Built automated Playwright/JavaScript regression tests for insurance quoting workflows across 40+ states.
- Developed 12 functional tests validating state-dependent business logic, catching data-accuracy issues pre-release.
- Diagnosed and resolved recurring Jenkins CI pipeline failures across 10 groups, improving test reliability.
- Contributed 11 pull requests to a shared end-to-end test automation framework, improving test coverage.

### Dell Technologies, Software Engineer Intern

**May 2025 - Aug 2025**

- Built an AI platform recommending regression tests for reported software defects, using semantic similarity search.
- Built a REST pipeline pulling 5,000+ test cases & 1,400+ Jira defects, cutting retrieval time ~83% via multithreading.
- Developed a full-stack app (React & Flask+FastAPI) delivering AI test recommendations to iDRAC quality engineers, saving ~15 minutes per defect across 1,400+ Jira defects.
- Achieved 89% test case recommendation accuracy and reduced manual test case selection time to seconds.
- Presented results to engineering leadership, securing buy-in to expand the solution across Dell Quality Eng teams.

### TL James Lab - Grambling, Software Lead

**September 2024 - May 2025**

- Built an Arduino-based sensor payload capturing atmospheric temperature & pressure at ~100,000ft high.
- Optimized C firmware for I<sup>2</sup>C/SPI peripherals, reducing sampling latency by 20% and ensuring reliability.
- Resolved 30+ software/hardware issues by debugging with logic analyzers, and protocol analyzers for SPI/I<sup>2</sup>C.
- Led a 5-member team through the design lifecycle: schematics, PCB assembly, calibration, and field deployment.

### KNUST Engineering Lab, FullStack Engineering Intern

**August 2023 - November 2023**

- Built a circuit design and simulation app used by 1500+ engineering students, enabling visual schematic creation.
- Modeled 25+ circuit components (transistors, logic gates, passives) as an object-oriented TypeScript framework.
- Developed a WebGL-based schematic canvas (PixiJS) with real-time wire and node detection for live circuit design.
- Built REST APIs connecting the backend to PySpice/ngspice, enabling accurate SPICE-based circuit simulation.

## SKILLS AND PROFICIENCIES

- **Programming languages:** Python, Java, JavaScript, C, C++, Flutter, TypeScript.
- **Experience with:** MongoDB, SQL, REST API, Flask, Node.js, React, HTML, CSS, Docker, GitHub Actions, Angular

## PROJECTS

### Intellijobs

- Built a full-stack job platform (React, Flutter, FastAPI, PostgreSQL) serving 22 users with cross-ATS job aggregation.
- Engineered a pipeline that surfaces job listings from 4 ATS platforms, aggregating opportunities in near real-time.
- Implemented rules-based ranking, location-aware filtering & Telegram alerts, reducing time-to-apply for users.
- Backed the platform with 1,800+ unit, integration, and e2e tests across pytest, Flutter, Vitest, and Playwright.

### Orbit-sh: A Modular Unix Shell with Systems Integration

[github.com/henryagyare/orbit-sh](https://github.com/henryagyare/orbit-sh)

- Built a Unix shell in C supporting command parsing, foreground/background execution, & built-ins (cd, exit).
- Implemented two-stage pipelines (cmd1 | cmd2) using pipe() and dup2() for inter-process I/O redirection.
- Prevented zombie processes via a SIGCHLD handler with non-blocking waitpid(), ensuring child process reaping.

### BrillaAI: STEM Questions Synthesis Platform

[github.com/henryagyare/BrillaAi-Questions-Synthesis](https://github.com/henryagyare/BrillaAi-Questions-Synthesis)

- Developed an AI system that generated 15,600+ STEM practice questions by fine-tuning LLaMA-3 8B model.
- Used gradient checkpointing and 8-bit optimizers to improve training efficiency on a single Tesla T4 GPU.
- Built ETL pipelines to clean, structure, & validate 15,600+ records, enabling reproducible quality evaluation.